# Practical-6 OUTPUT

Initialization:

```
  GNU nano 7.2                                          dining.c *
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define N 5   // Number of philosophers

sem_t chopstick[N];

void* philosopher(void* num) {
    int id = *(int*)num;

    while (1) {
        // Thinking
        printf("Philosopher %d is thinking\n", id);
        sleep(1);

        // Pick left chopstick
        sem_wait(&chopstick[id]);
        printf("Philosopher %d picked left chopstick\n", id);

        // Pick right chopstick
        sem_wait(&chopstick[(id + 1) % N]);
        printf("Philosopher %d picked right chopstick\n", id);

^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo       M-A Set Mark
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo       M-6 Copy
```

```
  GNU nano 7.2                                          dining.c
        // Pick left chopstick
        sem_wait(&chopstick[id]);
        printf("Philosopher %d picked left chopstick\n", id);

        // Pick right chopstick
        sem_wait(&chopstick[(id + 1) % N]);
        printf("Philosopher %d picked right chopstick\n", id);

        // Eating
        printf("Philosopher %d is eating\n", id);
        sleep(2);

        // Put down chopsticks
        sem_post(&chopstick[id]);
        sem_post(&chopstick[(id + 1) % N]);

        printf("Philosopher %d put down chopsticks\n", id);
    }
}

int main() {
    pthread_t thread[N];
    int id[N];

    // Initialize semaphores

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```

```
  GNU nano 7.2                                          dining.c *
}

int main() {
    pthread_t thread[N];
    int id[N];

    // Initialize semaphores
    for (int i = 0; i < N; i++) {
        sem_init(&chopstick[i], 0, 1);
    }

    // Create philosopher threads
    for (int i = 0; i < N; i++) {
        id[i] = i;
        pthread_create(&thread[i], NULL, philosopher, &id[i]);
    }

    // Join threads
    for (int i = 0; i < N; i++) {
        pthread_join(thread[i], NULL);
    }

    return 0;
}

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^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo       M-6 Copy
```

Output:

```
aditya@aditya-HP-Laptop-15-db0xxx: ~/dining_philosopher
aditya@aditya-HP-Laptop-15-db0xxx:~$ mkdir dining_philosopher
cd dining_philosopher
aditya@aditya-HP-Laptop-15-db0xxx:~/dining_philosopher$ nano dining.c
aditya@aditya-HP-Laptop-15-db0xxx:~/dining_philosopher$ gcc dining.c -o dining -lpthread
aditya@aditya-HP-Laptop-15-db0xxx:~/dining_philosopher$ ./dining
Philosopher 0 is thinking
Philosopher 1 is thinking
Philosopher 2 is thinking
Philosopher 3 is thinking
Philosopher 0 picked left chopstick
Philosopher 1 is thinking
Philosopher 2 picked left chopstick
Philosopher 3 is thinking
Philosopher 4 picked rtft chopstick
Philosopher 1 is eating
Philosopher 2 picked rigt chopstick
Philosopher 3 is thinking
Philosopher 2 picked lef chopstick
Philosopher 1 is thinking
Philosopher 4 put down chopsticks
Philosopher 3 is eating
Philosopher 2 picked let chopstick
Philosopher 1 is thinking
Philosopher 2 is thinking
Philosopher 2 picked rigt chopstick
Philosopher 4 is thinking
Philosopher 3 picked a chopstick
Philosopher 1 is thinking
Philosopher 2 is thinking
Philosopher 2 put down et chopstick
Philosopher 4 is thinking
aditya@aditya-HP-Laptop-15-db0xxx:~/dining_philosopher$
```